

Enhancing One Health Through Wastewater-Based Surveillance: Development and Validation of a Multiplex ddPCR Tool to Differentiate Human and Livestock Contributions

Andrea Gobbo^{a,1} , Marie-Alice Fraiture^{a,1}, Laura Van Poelvoorde^a ,
Sigrid C.J. De Keersmaecker^a , Bavo Verhaegen^b, Cristina Garcia-Graells^b,
Koenraad Van Hoorde^b, Hadrien Maloux^c, Veronik Hutse^c, Pieter-Jan Ceysens^{d,2},
Nancy Roosens^{a,2,*}

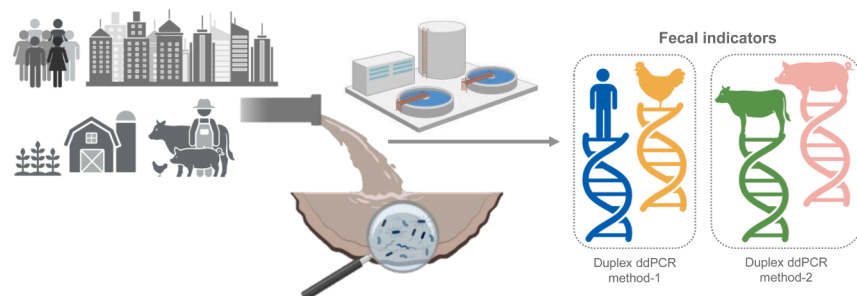
^a Transversal activities in Applied Genomics, Sciensano, Brussels, Belgium

^b Foodborne pathogens, Sciensano, Brussels, Belgium

^c Epidemiology of infectious diseases, Sciensano, Brussels, Belgium

^d Bacterial diseases, Sciensano, Brussels, Belgium

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Fecal indicator
Wastewater
ddPCR
Detection
Absolute quantification
Surveillance

ABSTRACT

Wastewater surveillance has emerged as valuable tool for public health, with notable successes in monitoring SARS-CoV-2 and, more recently, antimicrobial resistance (AMR) genes. Traditionally focused on human populations, this approach also holds significant promise for One Health surveillance. However, the extent to which livestock contributes to the microbial and genetic content detected in wastewater, compared to human sources, remains unclear. This gap in knowledge hampers our ability to normalize and accurately interpret wastewater surveillance data, particularly when integrating AMR gene signals. To address this, we developed and validated

* Corresponding author at: Rue Juliette Wytsman 14, 1050 Ixelles, Belgium.

E-mail addresses: Andrea.Gobbo@sciensano.be (A. Gobbo), Marie-Alice.Fraiture@sciensano.be (M.-A. Fraiture), Laura.VanPoelvoorde@sciensano.be (L. Van Poelvoorde), Sigrid.DeKeersmaecker@sciensano.be (S.C.J. De Keersmaecker), Bavo.Verhaegen@sciensano.be (B. Verhaegen), MariaCristina.GarciaGraells@sciensano.be (C. Garcia-Graells), Koenraad.VanHoorde@sciensano.be (K. Van Hoorde), Hadrien.Maloux@sciensano.be (H. Maloux), Veronik.Hutse@sciensano.be (V. Hutse), Pieter-Jan.Ceysens@sciensano.be (P.-J. Ceysens), Nancy.Roosens@sciensano.be (N. Roosens).

¹ equal first-author contribution.

² equal last-author contribution.

<https://doi.org/10.1016/j.watres.2025.125024>

Received 8 October 2025; Received in revised form 14 November 2025; Accepted 21 November 2025

Available online 23 November 2025

0043-1354/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

two duplex droplet digital PCR methods targeting fecal indicators specific to humans and key livestock species. These methods target cytochrome b of mitochondrial DNA specific to chickens and *Bacteroides* DNA specific either to humans, cows or pigs. Performance was evaluated against international standards to ensure reliability and harmonization across laboratories. A fecal DNA bank of 41 samples, collected from various human and animal sources, was used to retain only specific methods. Using serial dilutions of synthetic control plasmids carrying the targeted fecal indicator sequences, sensitivity was successfully assessed, with limits of detection below 10 copies. Applicability was demonstrated using wastewater samples spiked with livestock wastes. A one-year pilot monitoring was also conducted at two Belgian wastewater treatment plants, in regions with contrasting livestock populations. This study represents an essential step toward enhancing wastewater surveillance by integrating robust key species-specific fecal indicators. It lays the foundation for improved surveillance of livestock-associated outbreaks and the spread of AMR within a One Health framework.

1. Introduction

Wastewater-based surveillance (WBS) has become an essential tool for public health, particularly during the COVID-19 pandemic enabling early detection of viral trends at the community level, often before individuals showed symptoms or underwent testing (Rainey et al., 2023). A key factor contributing to this success was the use of fecal indicators, which helped normalizing pathogen concentrations in raw wastewater (Rainey et al., 2023; Maal-Bared et al., 2023). Among these, Pepper Mild Mottle Virus (PMMoV), an RNA-based fecal indicator specific to human waste, is widely used to adjust variations in sample composition, especially when wastewater flow rate data are missing or inconsistent (Rainey et al., 2023; Langeveld et al., 2023). This normalization strategy is also nowadays recommended by health authorities, including the U.S. Centers for Disease Control and Prevention (CDC) through its national wastewater surveillance system (Rainey et al., 2023).

While WBS has traditionally focused on human population monitoring, its application in livestock settings is gaining momentum, particularly for the early detection of disease outbreaks on farms. Livestock can serve as both reservoirs and sentinels for emerging pathogens, positioning it as a potential critical element for public health surveillance. This approach aligns with the One Health framework, which emphasizes the interconnectedness of human, animal, and environmental health (Parkins; Ahmed et al., 2024; Halliday et al., 2007; Bird Flu Found in, 2025; Xiao and Zhang, 2023; National Academies of Sciences, Engineering, and Medicine; Health, and Medicine Division, 2025). However, despite its potential, our understanding of the extent to which livestock relatively contributes to the microbial and genetic content in wastewater, compared to human sources, remains limited. This knowledge gap complicates the interpretation of wastewater surveillance data, especially when analyzing antimicrobial resistance (AMR) gene signals that may originate from both human and animal sources (National Academies of Sciences, Engineering, and Medicine; Health, and Medicine Division, 2025; Niegowska et al., 2021; Gholizadeh et al., 2023; La Rosa et al., 2025; Tiwari et al., 2022). Indeed, although wastewater treatment plants (WWTPs) are generally designed to collect primarily human fecal waste, such as in EU (Directive - UE - 2024/3019 - EN - EUR-Lex, 2025), it is important to recognize that wastewater does not always exclusively reflect human fecal inputs, as it may also contain waste from wild and domestic animals (Rainey et al., 2023). For instance, fecal contributions from livestock may result from agricultural runoff or accidental contamination from farms or slaughterhouses (Burkholder et al., 2007). While such contributions may have a limited impact when monitoring human-specific pathogens like SARS-CoV-2, they become substantially more relevant in surveillance efforts focused, for instance, on zoonotic diseases or AMR, particularly within the One Health framework (Argudín et al., 2017).

Recent findings emphasized the importance of integrating livestock-specific fecal indicators into WBS to strengthen public health preparedness. For example, in 2024 in the United States, Avian H5N1 Influenza virus was detected through animal-pathogen WBS several days before a nearby dairy cattle outbreak was confirmed (Increasing the Utility of, 2024). Such early-warning capabilities are also becoming

increasingly pertinent for AMR monitoring, especially in light of recent policy developments such as the European Commission's proposed revision of the Urban Wastewater Treatment Directive which mandates AMR genes surveillance at the European level (Gobbo et al., 2024; Proposal for a revised Urban, 2024). However, it remains unclear whether AMR genes detected in wastewater originate exclusively from human population or whether livestock also contributes significantly, particularly at certain WWTPs. This question is especially relevant given that chickens, cows and pigs, together accounting for approximately 94 % of global food animal production, represent a major source of fecal waste (Ardakani et al., 2024).

Consequently, to ensure accurate interpretation of human-related WBS data and facilitate livestock outbreak monitoring through WBS, the design of fecal indicators capable of differentiating between human and animal waste inputs is essential. The systematic use of human and animal fecal indicators at each WWTP is crucial for advancing public health surveillance by clarifying to what extend the observed WBS data are primarily reflecting human and/or animal populations (Sánchez-Alfonso et al., 2020; Kildare et al., 2007). To date, a variety of nucleic acid regions derived from bacteria, bacteriophages, and viruses have been proposed as fecal indicators and some have been occasionally tested in wastewater (Sinton et al., 1998). Commonly used fecal indicators like bacterial species belonging to *Escherichia coli* and enterococci are useful for assessing overall fecal contamination levels (Tiwari et al., 2022; Callejas et al., 2024; The European Union, 2025), and interesting to have an idea of the total fecal contamination level. However, they also exhibit weak host specificity, thereby limiting their effectiveness in accurately tracking the host origin of fecal contamination in wastewater samples. In contrast, other fecal indicators, like enteric viruses, bacteriophages (Rector et al., 2023) and anaerobic bacteria belonging to the genus *Bacteroides* (D'Aoust et al., 2021) are more promising in reason of their higher host specificity. Furthermore, although microbial-based fecal indicators are more commonly used, host mitochondrial DNA (mtDNA) (Tanvir Pasha et al., 2019; Caldwell and Levine, 2009) has also been identified as a valuable species-specific marker for fecal contamination (Monteiro et al., 2021). However, despite the variety of currently reported fecal indicators, no consensus has yet been established for an optimal normalization approach in WBS (Rainey et al., 2023).

Nowadays, PCR-based methods are widely employed in WBS, and, within this context, droplet digital PCR (ddPCR) has gained increasing popularity due to its ability to detect and absolutely quantify targets without the need for reference materials, as well as its high tolerance to PCR inhibitors typically found in complex matrices like wastewater. Furthermore, ddPCR provides results in terms of the copy number of the targets in the tested sample (Gobbo et al., 2024; Van Poelvoorde et al., 2023; Van Poelvoorde et al., 2021). Despite its advantages, only few ddPCR methods have been developed to target fecal indicators in water samples. For human-specific fecal indicators, human cytochrome b (hCYTB484) and human-associated *Bacteroides* (HF183/BacR287, and *B. theta*) have been targeted in a variety of environmental samples, including sewage (Nshimiyimana et al., 2019), river water (Hart et al., 2023), marine and freshwater storm drain water (Cao et al., 2015),

wastewater from rivers (Zhu et al., 2020), and alpine lake outlets as well as snowmelt streams (Pendergraph et al., 2021). For animal-specific fecal indicators, ddPCR methods targeting either all mammals (*AllBac* tested in lake outlets and snowmelt streams) (Pendergraph et al., 2021), bovines (*CowM2* tested in river water (Hart et al., 2023)), or ruminants (*Rum2Bac* tested in river water (Hart et al., 2023)) have been reported. However, no panel of ddPCR methods targeting fecal indicators specific to key species for wastewater surveillance has been selected and validated in accordance with international performance standards.

To address this gap, we developed and validated two duplex ddPCR methods capable of detecting both human and key livestock (chickens, cows and pigs) fecal indicators. This was achieved using a recently established workflow designed to harmonize PCR-based wastewater monitoring, which has been applied to develop multiplex ddPCR methods for AMR gene surveillance in wastewater (Gobbo et al., 2024). These methods target cytochrome b (cytb) of mtDNA specific to chickens and *Bacteroides* DNA specific either to humans, cows or pigs. Method performance was evaluated in accordance with international standards to ensure reliability and harmonization across laboratories (Gobbo et al., 2024; Marchesi et al., 2015; European Commission. Joint Research Centre, 2023). The method specificity was first tested using an *in-house* fecal DNA bank of 41 fecal samples collected from various human, livestock, and other animal sources. The method sensitivity was then evaluated only for assays demonstrating high specificity using different concentration levels of the target sequences. The method applicability was further investigated using wastewater samples spiked with fecal materials from different livestock species. Finally, a preliminary pilot monitoring study was conducted at two WWTPs in Belgium, selected based on their contrasting regional livestock populations, to explore their relative contributions of human and livestock fecal sources to the wastewater microbial community. Within the One Health framework, this study supports therefore the advancement of WBS by introducing reliable host-specific fecal indicators.

2. Materials and methods

2.1. Fecal DNA bank collection

To construct the *in-house* fecal DNA bank used in this study, a total of 41 different fecal samples were collected using the DNA/RNA Shield™ Fecal Collection Tubes (Zymo Research, USA) (Table S1). For the five human source samples, feces were collected from anonymized individuals by ProDigest (Ghent, Belgium), in accordance with ethical approval from the University Hospital Ghent. For the livestock source samples, feces were sampled from seven chickens, eight cows and nine pigs across five different Belgian farms, respectively. In addition, feces were sampled from a range of other animals, including one cat, three dogs, one pigeon, one pony, one donkey, one horse, one rabbit, one sheep and one duck. A mixed sample of feces from ducks and goose sharing the same habitat was also included.

DNA was extracted from each fecal sample using the Quick-DNA™ HMW MagBead Kit (Zymo Research, USA) following an *in-house* adapted protocol (Table S2) (Gand et al., 2023; Quick, 2019). DNA concentrations were then measured using the Qubit 4 Fluorometer (Thermo Fisher Scientific, Belgium).

2.2. ddPCR assays

Each ddPCR reaction (20 µl) contained 2x ddPCR Supermix for probes (no dUTP) (Bio-Rad, Belgium), 900 nM of each primer (IDT, Belgium) and 250 nM of each probe (IDT, Belgium) (Table 1). Reactions were loaded with 70 µl of droplet generation oil for probes into a DG8™ cartridge (Bio-Rad, Belgium) and placed in a QX200™ droplet generator (Bio-Rad, Belgium). PCR amplification was performed on 40 µl of the generated droplets using a T100™ thermal cycler (Bio-Rad, Belgium), with an initial 3-minute cycle at 25 °C, a 10-minute cycle at 95 °C (Taq polymerase activation), 40 cycles at 94 °C for thirty seconds (denaturation) and 58.5 °C for one minute (annealing-extension), and a final 10-minute cycle at 98 °C (Taq polymerase inactivation).

Table 1

Characteristics of the selected probe-based PCR assays for human and animal fecal indicators. The table lists the nine probed-based PCR assays selected for initial evaluation based on a literature review. For each assay, the targets, the oligonucleotides sequences, the amplicon sizes (in base pairs), and the corresponding references are provided. The complete sequences of the amplicons for each assay are provided in Table S3. The probe-based methods selected for further validation are indicated by an asterisk.

Targets	Oligonucleotides sequences (5' → 3')		Expected amplicon sizes	Method references
Human <i>Bacteroides</i> (HF183BacR287)*	Forward	ATCATGAGTTCACATGTCCG	126 bp	(Green et al., 2014)
	Probe	FAM-CTAATGGAACGCATCCC-MGB-Eclipse		
	Reverse	CTTCCTCTCAGAACCCCTATCC		
Human crAssphage (CPQ 056)	Forward	CAGAAGTACAAACTCCTAAAAACGTAGAG	126 bp	(Sabar et al., 2022)
	Probe	HEX-AATAACGATTTACGTGATGTAAC-MGB-Eclipse		
	Reverse	GATGACCAATAACAAGCCATTAGC		
Chicken <i>Bacteroides</i>	Forward	AATATTGGTCAATGGGCGAGAG	104 bp	(Kobayashi et al., 2013)
	Probe	FAM-TCCTTCACGCTACTTGG-MGB-Eclipse		
	Reverse	CACGTAGTGTCCGTTATTCCCTTA		
Chicken cytb of mtDNA*	Forward	TAGCCATGCACTACACAGCAGACA	103 bp	(Schill and Mathes, 2008)
	Probe	HEX-ACTTGCCGGAACGTACAATACGGCT-3IABkFQ		
	Reverse	TTTGCGTGGAGATTCCGGATGAGT		
Chicken cytb 2 of mtDNA	Forward	AAATCCCAACCCCTACTAAAAATAAT	263 bp	(Zhuang et al., 2017)
	Probe	FAM-ACAACCTCCCTAATCGACCT-MGB-Eclipse		
	Reverse	CAGATGAAGAAGAAATGAGGCG		
Chicken NADH dehydrogenase subunit 5 (ND5)	Forward	ACCTCCCCAACTAGC	172 bp	(Zhuang et al., 2017)
	Probe	FAM-TCAACCCATGCCTTCTT-MGB-Eclipse		
	Reverse	TTGCCAATGGTTAGGCAGGAG		
Chicken <i>Brevibacterium avium</i> (LA35)	Forward	ACCGGATACGACCATCTGC	571 bp	(Schiaffino et al., 2020)
	Probe	FAM-CAGCAGGGAAGAAGCCTTCGGGTGACGGTA-BHQ		
	Reverse	TCCCCAGTGTCACTCACAGC		
Cow <i>Bacteroides</i> *	Forward	GGATTACAGCCCTACGGGTTTTA	101 bp	(Lee et al., 2010)
	Probe	HEX-AAGAGTAATGTGCACTACGTG-3IABkFQ		
	Reverse	GGAGTTAGCCGATGCTTATTCATATAA		
Pig <i>Bacteroides</i> *	Forward	GCATGAATTAGCTTGCTAAATTTGAT	117 bp	(Gourmelon et al., 2010)
	Probe	FAM-TCCACGGGATAGCC-MGB-Eclipse		
	Reverse	ACCTCATACGGTATTAATCCGC		

Raw data were acquired with a QX200™ droplet reader (Bio-Rad, Belgium) and analyzed using QuantaSoft software version 1.7.4.0917 (Bio-Rad, Belgium), with the threshold manually set between positive and negative droplets. Assays yielding fewer than three positive droplets were considered negative (Sánchez-Alfonso et al., 2020; Kildare et al., 2007), except for the sensitivity assessment of the two duplex ddPCR methods where all values were included. A No Template Control (NTC) was included in each assay. Moreover, for each ddPCR method, 100 copies of a positive control, consisting of a synthetic plasmid (GeneCust, France) carrying a single copy of the corresponding PCR amplicon sequence, was used (Table S3).

2.3. Selection of ddPCR methods targeting fecal indicators

Nine probe-based PCR methods targeting fecal indicators for humans and livestock (chickens, cows, and pigs) in wastewater were selected from previous studies (Lee et al., 2010; Green et al., 2014; Schill and Mathes, 2008; Gourmelon et al., 2010; Sabar et al., 2022; Kobayashi et al., 2013; Zhuang et al., 2017; Schiaffino et al., 2020).

Each of these probe-based PCR methods was initially tested individually using ddPCR technology in duplicate, with 10 ng of DNA originating from fecal samples of different human, chicken, cow, pig, duck and goose sources (Table S4).

2.4. Multiplex ddPCR assay design and performance assessment

Based on preliminary simplex ddPCR tests (Table S4), four ddPCR methods specific to either human, chicken, cow or pig fecal sources were selected. With the aim of designing two duplex ddPCR methods, all oligonucleotides from the selected ddPCR methods were evaluated *in-silico* using the IDT OligoAnalyzer tool (Owczarzy et al., 2008) to assess potential heterodimer interactions, prioritizing the minimal deltaG value (Table S5). On this basis, the human-specific and chicken-specific methods were combined into one duplex ddPCR (Duplex-1), while the pig-specific and cow-specific methods were combined into a second duplex ddPCR (Duplex-2). Similarly to ddPCR methods recently validated for global AMR wastewater surveillance, the performance of these two duplex ddPCR methods was then assessed, including evaluations of specificity and sensitivity, in accordance with an international standard guidance document, being the Minimum Performance Requirements (MPR) for GMO analysis of European Network of GMO Laboratories (ENGL) (Gobbo et al., 2024; Marchesi et al., 2015; European Commission. Joint Research Centre, 2023; Guidance documents, 2024).

For the specificity assessment, 10 ng of DNA from positive and negative materials from the *in-house* fecal DNA bank (see section 2.1) were tested in duplicate for each duplex ddPCR method (Table S1).

For the symmetric sensitivity assessment, each duplex ddPCR method was tested using serial dilutions of the corresponding synthetic control plasmid (Table S3) mixtures. Six dilution points were prepared for each synthetic control plasmid, corresponding to 20, 10, 5, 2, 1, and 0.1 estimated target copies. Each dilution point was tested in twelve replicates. The limit of detection (LOD₉₅ %), representing the number of target copies needed to achieve a 95 % probability of detection (POD), was calculated using the QuoData web application (Uhlig et al., 2015) (Table S6).

In addition, the asymmetric sensitivity, where one target is tested at a low concentration in the presence of high concentrations of the other target, and vice versa, was tested for each duplex ddPCR method (Guidance documents, 2024). To this end, for each duplex ddPCR method, two different mixtures of synthetic control plasmids were prepared: (i) one containing 20 000 estimated target copies of Target 1 and 20 estimated target copies of Target 2, and (ii) one containing 20 estimated target copies of Target 1 and 20 000 estimated target copies of Target 2. Each mixture was tested in twelve replicates.

2.5. Applicability assessment

The applicability of the two duplex ddPCR methods was assessed using a wastewater sample from a Belgian WWTP (WWTP-A). This sample was analyzed both in its original form and after being artificially spiked with fecal materials from either chicken, cow, or pig sources (from the fecal DNA bank). Specifically, 10 ml of the wastewater sample was mixed with 0.1 g of either Chicken-7, Cow-7, or Pig-6 manure. For cow and pig solutions, 100 µl were diluted in 10 ml of wastewater. DNA was extracted from these wastewater samples using the Nanotrap® Microbiome A; 10 ml Manual Protocol with MagMAX™ - Wastewater Ultra Nucleic Acid Isolation Kit (Ceres Nano, USA). DNA concentrations were then measured using the Qubit 4 Fluorometer (Thermo Fisher Scientific, Belgium), and 1.5 µl of DNA per PCR reaction was tested in duplicate for each duplex ddPCR method.

2.6. Pilot monitoring

A preliminary pilot monitoring study was conducted using the two developed duplex ddPCR methods on an *in-house* wastewater sample collection established from two Belgian WWTPs located in regions with differing livestock populations: WWTP-A (low livestock populations) and WWTP-B (high livestock populations) (Table S7). The samples were collected weekly over a one-year period (April 2024 to March 2025), as part of the National wastewater-based epidemiology surveillance program, using 24-hour time-proportional and volume-proportional auto-samplers. After the collection, the samples were transported on the same day at 4 °C, then either stored directly at −20 °C or processed for DNA/RNA extraction before storage at −80 °C. Before the DNA/RNA extraction, 50 ml of wastewater from each sample was concentrated via ultrafiltration using Centricon Plus-70 Centrifugal filters (100 kDa). DNA/RNA was then extracted using the Maxwell® RSC PureFood GMO and Authentication Kit (Promega, USA) (Janssens et al., 2022). For each duplex ddPCR method, 1.5 µL of DNA/RNA was tested in duplicate.

3. Results and discussion

3.1. Selection and design of the host-specific fecal indicator duplex ddPCR methods

Based on previous studies, nine probe-based PCR methods, targeting fecal indicators specific to human and livestock (chickens, cows and pigs) sources in wastewater, were identified to be transferred to a ddPCR assay (Table 1). As an initial step in method assessment, an *in-silico* specificity analysis is typically performed. However, evaluating the specificity of fecal indicator methods using *in-silico* analysis has proven inadequate due to fundamental limitations in conventional sequence databases hosted for instance by NCBI. These databases are indeed primarily organized to classify sequences by species rather than by group of species associated to a certain host organism. The latter crucial metadata are often missing, incomplete, or ambiguous. In addition, when such metadata are available, they rarely clarify whether a microorganism is host-specific or capable of colonizing *multiple species*. This lack of resolution poses a significant challenge when attempting to identify microbial species that can serve as reliable indicators of fecal contamination from specific hosts in wastewater. For example, certain bacterial species like *Bacteroides fragilis* are considered potential indicators (Niestępski et al., 2020), but their reliability depends on their exclusive or predominant presence in the feces of a particular host species.

An alternative strategy for *in-silico* specificity assessment involved analyzing publicly available microbiome datasets from platforms such as the NCBI Sequence Read Archive (SRA). However, this approach also proved ineffective. Although such resources are valuable, they often lack sufficient sequencing depth in genomic regions critical for target identification, particularly in the case of non-human species and less-studied animal species.

Given these limitations to perform an *in-silico* specificity evaluation, only experimental investigation was deemed appropriate, and the most effective approach consisted of establishing an *in-house* fecal DNA bank. This entailed collecting and extracting DNA directly from fecal samples originated from several and various well-identified human and animal hosts. It provided a controlled host-specific reference for validating fecal indicator methods and served as a robust foundation for assessing the specificity of molecular assays.

The nine identified probe-based PCR methods were therefore first experimentally tested in simplex using ddPCR (Table 1, Table S4). This preliminary evaluation was conducted on a subset of 12 samples from the *in-house* fecal DNA bank, including different human and livestock sources (chicken, cow, pig), but also a few additional avian sources (duck and goose). For the pig-specific and cow-specific fecal indicator methods, as expected, a positive signal was observed only for DNA materials originated from pig and cow fecal samples, respectively (Table S4). Among the two different human-specific fecal indicator methods tested, only the human *Bacteroides* method showed positive signals for all human DNA fecal samples and negative signals for all tested non-

human DNA fecal samples (Table S4). Of the five different chicken-specific fecal indicator methods tested, only the cytb of mtDNA method presented positive signals for all chicken DNA fecal samples and negative signals for tested non-chicken DNA fecal samples, including other poultry species like duck and goose, allowing to exclude false positive signals from wild avian species (Table S4).

Based on these results, four ddPCR methods (Human *Bacteroides*, Chicken cytb of mtDNA, Cow *Bacteroides*, and Pig *Bacteroides*), which showed high specificity for their respective targets, were selected and then combined in two duplex ddPCR methods, based on the most optimal *in-silico* oligonucleotide compatibility (Table S5). More precisely, the human-specific and chicken-specific fecal indicator methods were combined into one duplex ddPCR (Duplex-1), while the pig-specific and cow-specific fecal indicator methods were combined into a second duplex ddPCR (Duplex-2). The performance of these duplex ddPCR methods was then assessed in accordance with an international standard guidance document, analogous to recent validations of ddPCR methods for global AMR wastewater surveillance (Gobbo et al., 2024; Marchesi et al., 2015; European Commission. Joint Research Centre,

Table 2

Experimental specificity results for the two duplex ddPCR assays across the *in-house* fecal DNA bank. Results, carried out in duplicate, are indicated by “+” for presence and “-” for absence of PCR amplification signals. Assays yielding fewer than three positive droplets were classified as negative (Table S1). A low positive signal, symbolized by “(+)”, was observed for one dog fecal sample (Dog-1) using the chicken cytb assay that directly targets host DNA. This low positive signal is likely due to the dog’s chicken-based diet (see section 3.2.1 and Table S8).

Sample description		Duplex-1 ddPCR method		Duplex-2 ddPCR method	
		Human <i>Bacteroides</i>	Chicken cytb of mtDNA	Cow <i>Bacteroides</i>	Pig <i>Bacteroides</i>
Fecal DNA	Human-1	+	-	-	-
	Human-2	+	-	-	-
	Human-3	+	-	-	-
	Human-4	+	-	-	-
	Human-5	+	-	-	-
	Chicken-1	-	+	-	-
	Chicken-2	-	+	-	-
	Chicken-3	-	+	-	-
	Chicken-4	-	+	-	-
	Chicken-5	-	+	-	-
	Chicken-6	-	+	-	-
	Chicken-7	-	+	-	-
	Cow-1	-	-	+	-
	Cow-2	-	-	+	-
	Cow-3	-	-	+	-
	Cow-4	-	-	+	-
	Cow-5	-	-	+	-
	Cow-6	-	-	+	-
	Cow-7	-	-	+	-
	Cow-8	-	-	+	-
	Pig-1	-	-	-	+
	Pig-2	-	-	-	+
	Pig-3	-	-	-	+
	Pig-4	-	-	-	+
	Pig-5	-	-	-	+
	Pig-6	-	-	-	+
	Pig-7	-	-	-	+
	Pig-8	-	-	-	+
	Pig-9	-	-	-	+
	Cat-1	-	-	-	-
	Dog-1	-	(+)	-	-
	Dog-2	-	-	-	-
	Dog-3	-	-	-	-
	Donkey	-	-	-	-
	Duck-1	-	-	-	-
	Duck/Goose-1	-	-	-	-
	Horse-1	-	-	-	-
	Pigeon-1	-	-	-	-
	Pony-1	-	-	-	-
	Rabbit-1	-	-	-	-
	Sheep-1	-	-	-	-
Control plasmid	Human <i>Bacteroides</i>	+	-	-	-
	Chicken cytb of mtDNA	-	+	-	-
	Cow <i>Bacteroides</i>	-	-	+	-
	Pig <i>Bacteroides</i>	-	-	-	+

2023; Guidance documents , 2024).

3.2. Performance assessment of the selected duplex ddPCR methods

3.2.1. Method specificity

The two designed duplex ddPCR assays were experimentally assessed for their specificity across the entire *in-house* fecal DNA bank of 41 samples collected from various human, livestock, and other animal sources (

Table 2). As expected, all fecal samples collected from five humans, seven chickens, eight cows, and nine pigs gave only a positive signal for their respective method, while fecal samples from non-target sources gave a negative signal. However, interestingly, among the three domestic canine fecal samples (Dogs 1–3), one yielded a weak positive signal using the chicken-specific fecal indicator method (Table 2).

This result was likely attributable to the dog's diet. Whereas microbial-based methods targeting host-specific *Bacteroides* are used for identifying fecal contaminations from human, cow, and pig sources, the chicken-specific fecal indicator method targets *cytb* of mtDNA is specific to chicken DNA, which may be present in trace amounts in the dog's feces due to the ingestion of chicken-containing food. Similar observations have been reported in previous studies (Schill and Mathes, 2008; Zhuang et al., 2017). To further investigate the potential dietary origin of the chicken-specific signal observed in one of the canine samples, a controlled feeding experiment was conducted with a fourth dog (Dog-4). The dog was fed an exclusive diet of either chicken or salmon for a period of two weeks. Fecal samples collected after the chicken-based diet showed a weak positive signal using the chicken-specific fecal indicator method, whereas samples collected after the salmon-based diet tested negative (Table S8). These results confirmed the hypothesized influence of diet. Furthermore, they underscore the importance of considering dietary sources when interpreting results from fecal indicator methods that target directly host DNA, as opposed to microbiome-based methods such as those targeting *Bacteroides*.

Based on all these results, given no false-positive and false-negative signals were observed, the two proposed duplex ddPCR methods were assessed specific.

3.2.2. Method sensitivity

For the symmetric sensitivity assessment, the two duplex ddPCR methods were tested on serial dilutions of each synthetic control plasmid mixture, ranging from 20 to 0.1 estimated target copies (

Table 3A, S3). A positive signal was detected in all twelve replicates up to 10 estimated target copies for the human-specific, chicken-specific, and cow-specific fecal indicator methods, and up to 5 estimated target copies for the pig-specific fecal indicator method. Based on all positive and negative signals observed for the twelve replicates at each dilution point, the LOD_{95 %} for each ddPCR method was calculated and found to be similar, ranging from 4 to 9 estimated target copies (Table 3A).

For the asymmetric sensitivity assessment, each duplex ddPCR method was tested using synthetic control plasmid mixtures containing either 20 000 estimated target copies of Target 1 and 20 estimated target copies of Target 2, and inversely (Table 3B). For each duplex ddPCR method, a positive signal was observed for all twelve replicates, with the measured target copy numbers being close to the expected values.

With LOD_{95 %} values below 25 estimated target copies, the developed duplex ddPCR methods were evaluated as sensitive and meeting international standard criteria (Marchesi et al., 2015; European Commission. Joint Research Centre, 2023; Guidance documents , 2024). In addition, these results demonstrate that the duplex ddPCR methods are not affected by the presence of highly abundant targets relative to the other. Given that human fecal contamination is expected to be predominant in WWTPs, these results also indicate that the livestock fecal indicator methods remain applicable in such settings, even if the associated fecal contaminations are present at trace levels.

Table 3

Experimental sensitivity results for the two duplex ddPCR assays. Amplification results are indicated by “+” for presence and “-” for absence.

(A) Symmetric sensitivity. Twelve replicates were tested for each estimated target copy. The number of positive replicates and the mean measured copy values (cp) and its standard deviation are provided, along with the calculated LOD _{95 %} (Table S6).				
Estimated target copies	Duplex-1		Duplex-2	
	Human <i>Bacteroides</i>	Chicken <i>cytb</i> of mtDNA	Cow <i>Bacteroides</i>	Pig <i>Bacteroides</i>
20	+, (12/12), (22.4 ± 8.0 cp)	+, (12/12), (17.6 ± 4.9 cp)	+, (12/12), (20.0 ± 3.8 cp)	+, (12/12), (18.2 ± 6.8 cp)
10	+, (12/12), (10.9 ± 3.7 cp)	+, (12/12), (9.1 ± 4.7 cp)	+, (12/12), (11.2 ± 3.7 cp)	+, (12/12), (9.2 ± 4.7 cp)
5	+, (10/12), (5.1 ± 5.9 cp)	+, (9/12), (3.1 ± 2.9 cp)	+, (11/12), (4.5 ± 3.1 cp)	+, (12/12), (5.4 ± 3.0 cp)
2	+, (10/12), (2.3 ± 1.6 cp)	+, (7/12), (1.7 ± 1.9 cp)	+, (9/12), (2.0 ± 1.9 cp)	+, (9/12), (1.7 ± 1.5 cp)
1	+, (4/12), (0.8 ± 1.0 cp)	+, (2/12), (0.4 ± 0.9 cp)	+, (4/12), (0.6 ± 1.0 cp)	+, (5/12), (0.6 ± 0.7 cp)
0.1	-, (0/12)	+, (1/12), (0.2 ± 0.5 cp)	-, (0/12)	+, (1/12), (0.1 ± 0.3 cp)
0	-, (0/12)	-, (0/12)	-, (0/12)	-, (0/12)
LOD _{95 %} (copies)	6	9	6	4
(B) Asymmetric sensitivity. Twelve replicates were tested for each estimated target copy, and the mean measured copy values (cp) and its standard deviation are provided.				
Duplex methods	Synthetic control plasmid mixtures	Target 1	Target 2	
Duplex-1, Human <i>Bacteroides</i> , + Chicken <i>cytb</i> of mtDNA	Human 20 000 cp + Chicken 20 cp	+ (17 648.3 ± 245.9 cp)	+ (24.3 ± 4.4 cp)	
	Human 20 cp + Chicken 20 000 cp	+ (22.5 ± 5.9 cp)	+ (19 561.7 ± 456.2 cp)	
Duplex-2 Cow <i>Bacteroides</i> + Pig <i>Bacteroides</i>	Cow 20 000 cp + Pig 20 cp	+ (20 141.7 ± 291.9 cp)	+ (26.0 ± 7.6 cp)	
	Cow 20 cp + Pig 20 000 cp	+ (19.6 ± 7.9 cp)	+ (19 103.3 ± 302.7 cp)	

3.3. Applicability assessment of the validated duplex ddPCR methods

Given that wastewater samples represent a complex matrix containing potential PCR inhibitors, the applicability of the two validated duplex ddPCR methods was evaluated using wastewater samples. The use of ddPCR technology is particularly advantageous due to its high tolerance to PCR inhibitors. This applicability assessment was conducted on a wastewater sample both in its original form and after being artificially spiked with an equivalent amount (in grams) of fecal wastes from either chicken, cow or pig (Table 4).

In the non-spiked wastewater sample, a positive signal associated to the human-specific fecal indicator was detected, consistent with the high population of the area from which the wastewater sample was collected. A weak positive signal for the chicken-specific fecal indicator was also observed. As previously demonstrated in a controlled feeding experiment with a dog (Table S8), a chicken-based diet can result in positive chicken mtDNA signal. Based on this finding, and given that chicken is a widely consumed food by humans (Bilans d'approvisionnement | Statbel, 2025), the detection of chicken mtDNA in wastewater where human inputs dominate may likely be attributed to the presence of dietary

Table 4

Experimental applicability evaluation of the two duplex ddPCR assays using a wastewater sample either non-spiked or artificially spiked with livestock fecal wastes. Amplification results are indicated by “+” for presence and “-” for absence in the first row, and the mean copies per ml of wastewater and its standard deviation, in the last row. Assays yielding fewer than three positive droplets were classified as negative. All results were confirmed through duplicate experiments.

Wastewater samples	ddPCR results expressed in copies/ml wastewater			
	Duplex-1		Duplex-2	
	Human <i>Bacteroides</i>	Chicken cytb of mtDNA	Cow <i>Bacteroides</i>	Pig <i>Bacteroides</i>
Non-spiked	+	+	-	-
	(520 ± 7.1)	(19.3 ± 1.9)		
Spiked with chicken manure	+	+	-	-
	(103.3 ± 2.8)	(32 911.0 ± 70.7)		
Spiked with pig manure	+	+	-	+
	(533.3 ± 2.8)	(22 ± 1.1)		(180 333.3 ± 424.3)
Spiked with cow manure	+	+	+	-
	(510 ± 26.9)	(26 ± 3.9)	(9 166.7 ± 42.4)	

chicken residues in human waste. Nonetheless, due to the lack of dietary information from the human waste donors in this study, this interpretation remains a hypothesis supported by indirect evidence. Future controlled human dietary studies will consequently be valuable to confirm this hypothesis. In addition, no signal was reported for the cow- and pig-specific fecal indicators. The absence or low levels of livestock-specific signals were consistent with the low livestock populations in the area associated to the tested wastewater sample.

In all wastewater samples artificially spiked with chicken, cow or pig manures, a positive signal for the human-specific fecal indicator was consistently detected, as expected and in agreement with the result observed in the non-spiked wastewater sample. When spiked with chicken manure, only the chicken-specific fecal indicator yielded a positive signal among the livestock-specific markers tested. When spiked with cow or pig manures, a positive signal was respectively detected for the cow- and pig-specific fecal indicators, as well as, similarly to the non-spiked wastewater sample, a weak signal for the chicken-specific fecal indicator was reported. Interestingly, although equal amounts (in grams) of chicken, cow, and pig manure were used for spiking, the corresponding host-specific fecal indicators did not produce signals of comparable magnitude. This observation suggests that, for a similar mass input, the concentration of target fecal indicators may vary among host sources. Such differences likely reflect biological variability among individual hosts, including differences in gut microbiome composition, diet, and physiology, all of which influence the fecal indicator concentrations independently of fecal mass. This variability should be carefully considered when interpreting results from fecal source tracking analyses. Nevertheless, this variability does not compromise the interpretation of wastewater trends. The objective of this study was to estimate fecal source contributions at the population level rather than from individual samples. Because wastewater represents a composite mixture of numerous fecal inputs, individual variations are effectively averaged out, resulting in overall marker patterns that should remain stable and representative of the dominant human or animal sources.

Based on all these results, the successful applicability of the two validated duplex ddPCR methods to wastewater samples was demonstrated.

3.4. WBS pilot monitoring

Using the validated duplex ddPCR methods, a preliminary WBS pilot study was conducted on wastewater samples collected weekly over one

year from two different Belgian WWTPs (

Fig. 1). WWTP-A's catchment area covers a bigger human population and a smaller livestock population compared to WWTP-B (Table S7). WWTP-B was located in a rural area with relatively higher livestock density for Belgium. These two sites were chosen because they represent the most extreme conditions available within the Belgian wastewater monitoring program. No additional WWTPs suitable for the present study are currently available.

In wastewater samples from both WWTPs, a strong positive signal corresponding to the human-specific fecal indicator was observed. Notably, WWTP-A, with its larger associated urban population, showed a higher signal than WWTP-B.

In contrast to the human-specific fecal indicator, a weak or even nearly undetectable signal was measured for the chicken-, cow- and pig-fecal indicators in wastewater samples from both WWTPs (Table S9 A). Among these, the weak signal associated to the chicken-specific fecal indicator is likely reflecting dietary background sources from human population as previously reported (Fig. 1, Table S9 B). In addition, although weak signals of cow- and pig-specific fecal indicators are also observed, their intensity is higher in WWTP-B than in WWTP-A, which could be explained by the respective human and livestock population in the catchment areas (Fig. 1, Table S7).

These results, reflecting the specific Belgian setting, provide preliminary insights into the relative contributions of human and livestock fecal contamination in the wastewater from the investigated WWTPs. These results were in line with the distribution of human and livestock contributions in Belgium, where human inputs dominate even in rural area with agricultural activities. Additionally, as expected, livestock-specific fecal indicators presented a higher signal in the rural area than in the urban area, confirming the consistency of the analysis with the local land-use and population characteristics. These results also suggest that these two WWTPs used in this study are primarily useful for monitoring human populations rather than livestock. This is likely due to the high human populations in the catchment areas compared to livestock, and the design of the sewage network, which limits farm connections in accordance with the European regulations (Directive - UE - 2024/3019 - EN - EUR-Lex, 2025). Therefore, to effectively monitor livestock-related fecal contamination, the identification of other WWTPs, or analogous facilities, serving more agricultural or mixed-use will be needed. Future studies involving other environmental samples, such as those from farms or agricultural sites, could also be valuable. However, analyses of these samples should be conducted separately from wastewater samples in order to preserve the scope and interpretability of the present WWTP-focused pilot monitoring and to prevent conflating results.

4. Conclusion

This study presents the development and validation of two duplex ddPCR methods targeting DNA-based host-specific fecal indicators associated with human and key livestock (chicken, cow and pig) populations. These ddPCR methods, applicable to samples containing either a single fecal source or a mixture of multiple fecal sources, allow to facilitate raw WBS data interpretation and distinguish species-specific fecal source contributions in wastewater. For instance, this study is particularly relevant to support surveillance efforts focused on zoonotic diseases or AMR, particularly within the One Health framework. Quantifying host-specific fecal indicators provides useful information on the relative contributions of human and livestock fecal inputs in a given sample. Consequently, when an AMR gene is detected in a wastewater sample dominated by human fecal inputs, it is more likely to originate from human sources.

The performance of these duplex ddPCR methods was evaluated in accordance with international standard criteria. The specificity assessment confirmed that both duplex ddPCR methods detected their respective targets (Table 2). Interestingly, in contrast to host-specific

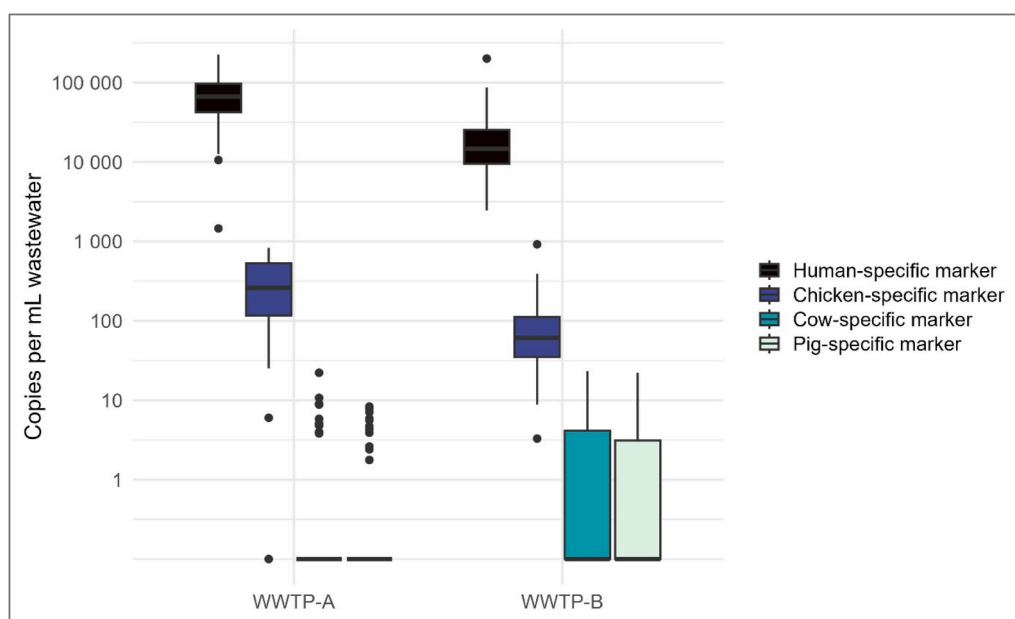


Fig. 1. Experimental results for the wastewater samples from WWTP-A and WWTP-B, collected weekly over 1 year, using the two duplex ddPCR assays. WWTP-A is located in an area with higher human population and lower livestock population compared to WWTP-B. Boxplot graph showing the ddPCR results for the 104 wastewater samples expressed in logarithmic scale. Assays yielding fewer than three positive droplets were classified as negative. All results were confirmed through duplicate experiments (Table S9 A).

Bacteroides-based methods (used for human, cow, and pig fecal indicators), the host-specific cytb of mtDNA-based method (used for chicken fecal indicator) may slightly be influenced by a chicken-based diet (Table S8). This specificity assessment was made possible by the use of an *in-house* fecal DNA bank, established to include samples from a wide range of several human, livestock, and other animal sources. Built from local Belgian sources, this DNA bank represents a valuable resource that adequately captures the diversity needed to assess method specificity within the scope of the present study on Belgian wastewater samples. The inclusion of samples from different breeds, diets, and geographical regions would further enhance the validation of the method's specificity. However, expanding the DNA bank to comprehensively represent global, or even only Europe-wide, diversity would require extensive additional sampling, coordination, and analytical efforts that go generally beyond the objectives and available resources of a single laboratory or even a single country. Moreover, the collection of large numbers of human fecal samples poses ethical challenges that further limit the feasibility of constructing a truly global DNA bank. Consequently, at the practical level, capturing the full extent of global fecal community diversity within a single DNA bank is highly challenging. Therefore, for effective, robust, and regionally relevant analyses, the DNA bank needs to be expanded by incorporating additional fecal DNA samples, particularly from local sources aligned with the specific study context and intended analytical scope (e.g., at the country level), since microbial communities and markers may vary geographically, as well as in accordance with regional ethical considerations. Following the specificity assessment, the symmetric sensitivity assessment demonstrated that the duplex ddPCR methods reliably detected low copy numbers of their respective targets, with LOD₉₅ % values ranging from 4 to 9 estimated target copies (Table 3A). Additionally, the asymmetric sensitivity analysis confirmed that the assays remained unaffected by the presence of highly abundant co-targets (Table 3B). Finally, the applicability assessment, using artificially spiked

wastewater samples, demonstrated that the developed duplex ddPCR assays successfully detected both human and livestock fecal indicators in wastewater (Table 4).

A preliminary WBS pilot monitoring study was then conducted using the two validated duplex ddPCR methods using wastewater samples collected over one year from two Belgian WWTPs with varying human and livestock populations. The results indicated a strong signal for the human fecal indicator and weak signals for the livestock fecal indicators, in a manner consistent with the expected population and agricultural populations. For the chicken fecal indicator, which directly targets chicken DNA, a weak signal was noticed compared to the other livestock indicators, probably due to local dietary habits. Based on this preliminary overview regarding the relative contributions of fecal contaminations from human and key livestock species in wastewater, it was concluded that livestock fecal waste has no major contribution on the wastewater composition from the two investigated WWTPs. However, these observations may not be universally applicable to all WWTPs, as wastewater composition depends on local conditions. Moreover, the potential contribution of wild fauna, particularly migratory birds, remains unknown, and in certain locations, their impact on the occurrence of specific zoonoses (e.g., avian influenza viruses) could be significant. Further method development and research are needed to assess this potential influence. To gain a more comprehensive understanding of human and livestock waste inputs in wastewater, a geographically expanded study covering multiple WWTPs in areas with different human and livestock populations will be needed. Additionally, beyond wastewater, the duplex ddPCR methods developed in this study could be valuable for other applications, including monitoring environmental water sources in both urban and agricultural areas, as well as supporting in-farm surveillance. Furthermore, based on new insights and emerging needs, additional ddPCR methods may be incorporated into the initially proposed set in order to target fecal indicators specific to other species of interest. Finally, as RNA-based fecal indicators, such as PMMoV, may

introduce additional variability due to differences in extraction efficiency and stability between RNA and DNA molecules, the use of DNA-based indicators were deemed more appropriate to ensure methodological consistency and accurate data interpretation in this study, which focused on the normalization of DNA targets. Nonetheless, RNA-based fecal indicators could serve as valuable complementary tools in future studies, integrating both RNA and DNA targets within a unified surveillance framework.

Ethics approval statement

The five human fecal samples were collected by ProDigest (Ghent, Belgium) from anonymized individuals according to the ethical approval of the University Hospital Ghent.

Patient consent statement

Not applicable.

Permission to reproduce material from other sources

Not applicable.

Clinical trial registration

Not applicable.

Funding

This research was funded by the Belgian ‘One Health’ National Action Plan on the Fight Against Antimicrobial Resistance (2020–2024) and by internal Sciensano funding.

CRedit authorship contribution statement

Andrea Gobbo: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Marie-Alice Fraiture:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Laura Van Poelvoorde:** Writing – review & editing, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Sigrid C.J. De Keersmaecker:** Writing – review & editing, Project administration, Methodology. **Bavo Verhaegen:** Writing – review & editing, Project administration, Methodology. **Cristina Garcia-Graells:** Writing – review & editing, Project administration, Methodology. **Koenraad Van Hoorde:** Writing – review & editing, Project administration, Methodology. **Hadrien Maloux:** Writing – review & editing, Project administration, Formal analysis, Data curation. **Veronik Hutse:** Writing – review & editing, Project administration, Formal analysis, Data curation. **Pieter-Jan Ceysens:** Writing – review & editing, Validation, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Nancy Roosens:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

Acknowledgments

Acknowledgments to Bram Bloemen, Gauthier Lequeue, Jolien D’aes, Mathieu Gand, Maud Delvoye, Michaël Kelchtermans, Nina Papazova, Yari Van Laere and the voluntary farmers for their

contributions in helping us establish the fecal DNA bank. Acknowledgments to Eliza Al Bitar for the collection of fecal samples from dog and the associated feeding experiments.

Data availability

Data will be made available on request.

References

- Rainey, A.L., Liang, S., Jr, J.H.B., Sabo-Attwood, T., Maurelli, A.T., 2023. A multistate assessment of population normalization factors for wastewater-based epidemiology of COVID-19. *PLOS ONE* 18, e0284370.
- Maal-Bared, R., Qiu, Y., Li, Q., Gao, T., Hruddy, S.E., et al., 2023. Does normalization of SARS-CoV-2 concentrations by Pepper Mild Mottle Virus improve correlations and lead time between wastewater surveillance and clinical data in Alberta (Canada): comparing twelve SARS-CoV-2 normalization approaches. *Sci. Total Environ.* 856, 158964.
- Langeveld, J., Schilperoord, R., Heijnen, L., Elsinga, G., Schapendonk, C.E.M., et al., 2023. Normalisation of SARS-CoV-2 concentrations in wastewater: the use of flow, electrical conductivity and crAssphage. *Sci. Total Environ.* 865, 161196.
- Parkins, M.D., Lee, B.E., Acosta, N., Bautista, M., Hubert, C.R.J., et al., 2024. Wastewater-based surveillance as a tool for public health action: sARS-CoV-2 and beyond. *Clin. Microbiol. Rev.* 37, e00103–22.
- Ahmed, W., Liu, Y., Smith, W., Ingall, W., Belby, M., et al., 2024. Leveraging wastewater surveillance to detect viral diseases in livestock settings. *Sci. Total Environ.* 931, 172593.
- Halliday, J.E.B., Meredith, A.L., Knobel, D.L., Shaw, D.J., Bronsvort, B.M., de, C., et al., 2007. A framework for evaluating animals as sentinels for infectious disease surveillance. *J. R. Soc. Interface* 4, 973–984.
- Bird Flu Found in Wastewater before reported outbreaks. <https://www.idsociety.org/news-publications-new/articles/2024/bird-flu-found-in-wastewater-before-reported-outbreaks/> (accessed 16 May 2025).
- Xiao, K., Zhang, L., 2023. Wastewater pathogen surveillance based on One Health approach. *Lancet Microbe* 4, e297.
- National Academies of Sciences, Engineering, and Medicine; Health and Medicine Division; Board on population health and public health practice; division on earth and life studies; water science and technology board; committee on community wastewater-based infectious disease surveillance. *Increasing the Utility of Wastewater-based Disease Surveillance for Public Health Action: A Phase 2 Report*. Washington (DC): National Academies Press (US). <http://www.ncbi.nlm.nih.gov/books/NBK610719/> (2024, accessed 5 September 2025).
- Niegowska, M., Sanseverino, I., Navarro, A., Lettieri, T., 2021. Knowledge gaps in the assessment of antimicrobial resistance in surface waters. *FEMS Microbiol. Ecol.* 97, fiab140.
- Gholizadeh, A., Khadani, M., Foroughi, M., Alizade Siuki, H., Mehrfar, H., 2023. Wastewater treatment plants: the missing link in global one-health surveillance and management of antibiotic resistance. *J. Infect. Public Health* 16, 217–224.
- La Rosa, M.C., Maugeri, A., Favara, G., La Mastra, C., Magnano San Lio, R., et al., 2025. The impact of wastewater on antimicrobial resistance: a scoping review of transmission pathways and contributing factors. *Antibiotics* 14, 131.
- Tiwari, A., Kurittu, P., Al-Mustapha, A.L., Heljanko, V., Johansson, V., et al., 2022. Wastewater surveillance of antibiotic-resistant bacterial pathogens: a systematic review. *Front. Microbiol.* 13, 977106.
- Directive - UE - 2024/3019 - EN - EUR-Lex. <https://eur-lex.europa.eu/eli/dir/2024/3019/oj/eng> (accessed 15 September 2025).
- Burkholder, J., Libra, B., Weyer, P., Heathcote, S., Kolpin, D., et al., 2007. Impacts of waste from concentrated animal feeding operations on water quality. *Environ. Health Perspect.* 115, 308–312.
- Argudín, M.A., Deplano, A., Meghraoui, A., Dodémont, M., Heinrichs, A., et al., 2017. Bacteria from animals as a pool of antimicrobial resistance genes. *Antibiotics (Basel)* 6, 12.
- Increasing the Utility of Wastewater-Based Disease Surveillance For Public Health Action: A Phase 2 Report. Washington, D.C.: National Academies Press. Epub ahead of print 6 December 2024. DOI: 10.17226/27516.
- Gobbo, A., Fraiture, M.-A., Van Poelvoorde, L., De Keersmaecker, S.C.J., Garcia-Graells, C., et al., 2024. Strategy to develop and validate digital droplet PCR methods for global antimicrobial resistance wastewater surveillance. *Water Environ. Res.* 96, e11145.
- Proposal for a revised Urban Wastewater treatment directive - European Commission. https://environment.ec.europa.eu/publications/proposal-revised-urban-waste-water-treatment-directive_en (accessed 8 April 2024).
- Ardakani, Z., Aragrande, M., Canali, M., 2024. Global antimicrobial use in livestock farming: an estimate for cattle, chickens, and pigs. *Animal* 18, 101060.
- Sánchez-Alfonso, A.C., Venegas, C., Díez, H., Méndez, J., Blanch, A.R., et al., 2020. Microbial indicators and molecular markers used to differentiate the source of faecal pollution in the Bogotá River (Colombia). *Int. J. Hyg. Environ. Health* 225, 113450.
- Kildare, B.J., Leutenegger, C.M., McSwain, B.S., Bambic, D.G., Rajal, V.B., et al., 2007. 16S rRNA-based assays for quantitative detection of universal, human-, cow-, and dog-specific fecal *bacteroides*: a bayesian approach. *Water Res.* 41, 3701–3715.
- Sinton, L.W., Finlay, R.K., Hannah, D.J., 1998. Distinguishing human from animal faecal contamination in water: a review. *N. Z. J. Mar. Freshwater Res.* 32, 323–348.

- Callejas, I.A., Kong, Y., Osborn, K., Hung, W.-C., Cira, M., et al., 2024. The influence of urbanization and water reclamation plants on fecal indicator bacteria and antibiotic resistance in the Los Angeles River watershed: a case study with complementary monitoring methods. *Sci. Total Environ.* 957, 177577.
- The European Union Summary Report on Antimicrobial Resistance in zoonotic and indicator bacteria from humans, animals and food in 2020/2021 - 2023 - EFSA Journal - Wiley Online Library. <https://efsa.onlinelibrary.wiley.com/doi/full/10.2903/j.efsa.2023.7867> (accessed 16 May 2025).
- Rector, A., Bloemen, M., Thijssen, M., Delang, L., Raymenants, J., et al., 2023. Monitoring of SARS-CoV-2 concentration and circulation of variants of concern in wastewater of Leuven, Belgium. *J. Med. Virol.* 95, e28587.
- D'Aoust, P.M., Mercier, E., Montpetit, D., Jia, J.-J., Alexandrov, I., et al., 2021. Quantitative analysis of SARS-CoV-2 RNA from wastewater solids in communities with low COVID-19 incidence and prevalence. *Water Res.* 188, 116560.
- Tanvir Pasha, A.B.M., Hinojosa, J., Phan, D., Lopez, A., Kapoor, V., 2019. Detection of human fecal pollution in environmental waters using human mitochondrial DNA and correlation with general and human-associated fecal genetic markers. *J. Water Health* 18, 8–18.
- Caldwell, J.M., Levine, J.F., 2009. Domestic wastewater influent profiling using mitochondrial real-time PCR for source tracking animal contamination. *J. Microbiol. Methods* 77, 17–22.
- Monteiro, S., Ebdon, J., Santos, R., Taylor, H., 2021. Elucidation of fecal inputs into the River Tagus catchment (Portugal) using source-specific mitochondrial DNA, HAdV, and phage markers. *Sci. Total Environ.* 783, 147086.
- Van Poelvoorde, L.A.E., Picalausa, C., Gobbo, A., Verhaegen, B., Lesenfants, M., et al., 2023. Development of a droplet digital PCR to monitor SARS-CoV-2 omicron variant BA.2 in wastewater samples. *Microorganisms* 11, 729.
- Van Poelvoorde, L.A.E., Gand, M., Fraiture, M.-A., De Keersmaecker, S.C.J., Verhaegen, B., et al., 2021. Strategy to develop and evaluate a multiplex RT-ddPCR in response to SARS-CoV-2 genomic evolution. *Curr. Issues Mol. Biol.* 43, 1937–1949.
- Nshimiyimana, J.P., Cruz, M.C., Wuertz, S., Thompson, J.R., 2019. Variably improved microbial source tracking with digital droplet PCR. *Water Res.* 159, 192–202.
- Hart, J.J., Jamison, M.N., McNair, J.N., Woznicki, S.A., Jordan, B., et al., 2023. Using watershed characteristics to enhance fecal source identification. *J. Environ. Manage.* 336, 117642.
- Cao, Y., Raith, M.R., Griffith, J.F., 2015. Droplet digital PCR for simultaneous quantification of general and human-associated fecal indicators for water quality assessment. *Water Res.* 70, 337–349.
- Zhu, K., Suttner, B., Pickering, A., Konstantinidis, K.T., Brown, J., 2020. A novel droplet digital PCR human mtDNA assay for fecal source tracking. *Water Res.* 183, 116085.
- Pendergraph, D.P., Ranieri, J., Ermatinger, L., Baumann, A., Metcalf, A.L., et al., 2021. Differentiating sources of fecal contamination to wilderness waters using droplet digital PCR and fecal indicator bacteria methods. *Wilderness Environ. Med.* 32, 332–339.
- Marchesi U., Mazzara M., Broll H., Giacomo M., Grohmann L., et al. *European Network of GMO Laboratories (ENGL) 2015 definition of minimum performance requirements for analytical methods of GMO testing*. 2015. Epub ahead of print 1 April 2015. DOI: 10.13140/RG.2.1.2060.5608.
- European Commission. Joint Research Centre. *Definition of Minimum Performance Requirements For Analytical Methods of GMO Testing. Part 2, European Network of GMO Laboratories : European Union Reference Laboratory For Genetically Modified Food and Feed*. LU: Publications Office. <https://data.europa.eu/doi/10.2760/63656> (2023, accessed 30 May 2024).
- Gand, M., Bloemen, B., Vanneste, K., Roosens, N.H.C., De Keersmaecker, S.C.J., 2023. Comparison of 6 DNA extraction methods for isolation of high yield of high molecular weight DNA suitable for shotgun metagenomics nanopore sequencing to detect bacteria. *BMC Genomics* 24, 438.
- Quick, J., 2019. The 'Three Peaks' faecal DNA extraction method for long-read sequencing. <https://www.protocols.io/view/the-39-three-peaks-39-faecal-dna-extraction-method-7rsh6e> accessed 16 May 2025.
- Lee, D.-Y., Weir, S.C., Lee, H., Trevors, J.T., 2010. Quantitative identification of fecal water pollution sources by TaqMan real-time PCR assays using Bacteroidales 16S rRNA genetic markers. *Appl. Microbiol. Biotechnol.* 88, 1373–1383.
- Green, H.C., Haugland, R.A., Varma, M., Millen, H.T., Borchardt, M.A., et al., 2014. Improved HF183 quantitative real-time PCR assay for characterization of Human fecal pollution in ambient surface water samples. *Appl. Environ. Microbiol.* 80, 3086–3094.
- Schill, W.B., Mathes, M.V., 2008. Real-time PCR detection and quantification of nine potential sources of fecal contamination by analysis of mitochondrial cytochrome b targets. *Environ. Sci. Technol.* 42, 5229–5234.
- Gourmelon, M., Caprais, M.P., Mieskin, S., Marti, R., Wéry, N., et al., 2010. Development of microbial and chemical MST tools to identify the origin of the faecal pollution in bathing and shellfish harvesting waters in France. *Water Res.* 44, 4812–4824.
- Sabar, M.A., Honda, R., Haramoto, E., 2022. CrAssphage as an indicator of human-fecal contamination in water environment and virus reduction in wastewater treatment. *Water Res.* 221, 118827.
- Kobayashi, A., Sano, D., Hatori, J., Ishii, S., Okabe, S., 2013. Chicken- and duck-associated Bacteroides-Prevotella genetic markers for detecting fecal contamination in environmental water. *Appl. Microbiol. Biotechnol.* 97, 7427–7437.
- Zhuang, F.-F., Li, H., Zhou, X.-Y., Zhu, Y.-G., Su, J.-Q., 2017. Quantitative detection of fecal contamination with domestic poultry feces in environments in China. *AMB Express* 7, 80.
- Schiaffino, F., Pisanic, N., Colston, J.M., Rengifo, D., Paredes Olortegui, M., et al., 2020. Validation of microbial source tracking markers for the attribution of fecal contamination in indoor-household environments of the Peruvian Amazon. *Sci. Total Environ.* 743, 140531.
- Owczarzy, R., Tataurov, A.V., Wu, Y., Manthey, J.A., McQuisten, K.A., et al., 2008. IDT SciTools: a suite for analysis and design of nucleic acid oligomers. *Nucleic Acids Res.* 36, W163–W169.
- Guidance documents | European Union reference laboratory for genetically modified food and feed (EURL GMFF). <https://gmo-crl.jrc.ec.europa.eu/guidance-documents> (accessed 21 August 2024).
- Uhlig, S., Frost, K., Colson, B., Simon, K., Mäde, D., et al., 2015. Validation of qualitative PCR methods on the basis of mathematical-statistical modelling of the probability of detection. *Accred. Qual. Assur.* 20, 75–83.
- Janssens, R., Hanoteaux, S., Maloux, H., Klammer, S., Laisnez, V., et al., 2022. SARS-CoV-2 surveillance in Belgian wastewaters. *Viruses* 14, 1950.
- Niestepski, S., Harnisz, M., Korzeniewska, E., Osińska, A., 2020. Markers specific to Bacteroides fragilis group bacteria as indicators of anthropogenic pollution of surface waters. *Int. J. Environ. Res. Public Health* 17, 7137.
- Bilans d'appvisionnement | Statbel. [h.ttps://statbel.fgov.be/fr/themes/agriculture-peche/bilans-dapprovisionnement#panel-12](https://statbel.fgov.be/fr/themes/agriculture-peche/bilans-dapprovisionnement#panel-12) (accessed 3 February 2025).